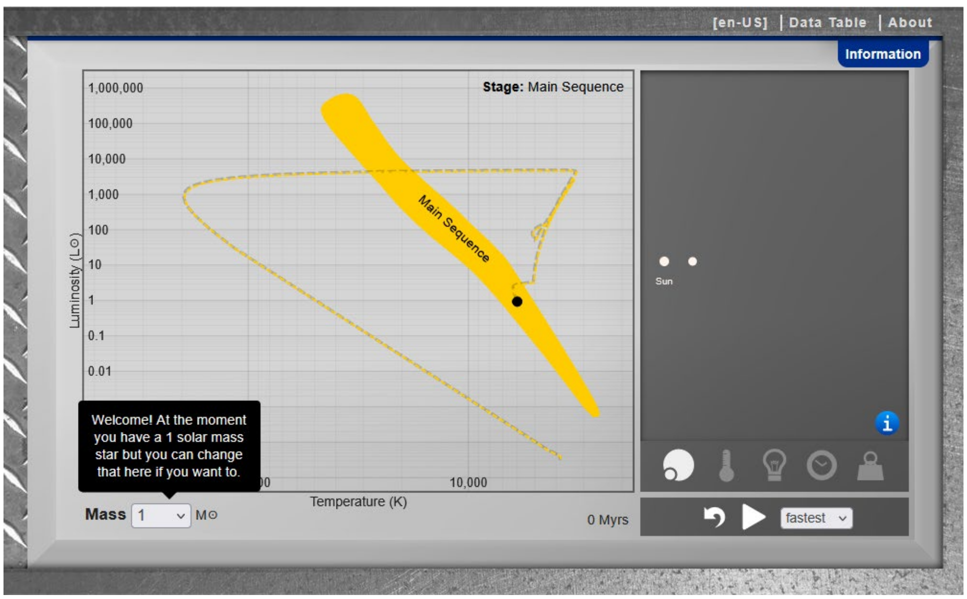
Homework – Star in a Box

Goals

In this activity, you will explore the different stages in the evolution of stars of various masses, both smaller than, the same size as, and larger than our Sun. You will see the correlation between the star sizes, ages, and position on the HR Diagram.

Using the Simulator

- Go to the [Star in a Box]() simulator.
- Click on the slider for **Normal/Advanced** to select **Advanced**.
- Click on **Open the lid**.



You will now see the simulator and it will look like the picture above. The main features are:

- Initially, it will be set to a star with the same mass as the Sun. This can be changed with the pulldown menu on the lower left.
- The main box on the left shows an HR Diagram and the lifecycle track of the Main Sequence and beyond for the star mass you have selected. You can also see the time going by, in millions of years, in the lower right.
- The right box shows the size of the star on the Main Sequence (leftmost circle) and the current size of the star based on where you are in the lifecycle (rightmost circle).
- At the bottom of the right box, there are two rows of controls to change what is displayed in the right box:
  - For the top row of controls (in left-to-right order)
    - *Size* comparison between the star on the Main Sequence and at the current stage of its life
    - *Surface temperature* of the star based on the current lifecycle stage

    - *Luminosity* of the star based on the current lifecycle stage
    - *Relative times* of each of the lifecycle stages of the star based on the mass
    - *Mass* of the star based on the current lifecycle stage
  - For the bottom row of controls (from left-to-right order)
    - *Resets* the simulator
    - *Starts* the animation tracing the selected star's lifecycle
    - Pulldown menu sets the *speed* of the animation

Solar Mass Star (our Sun)

First, we'll explore the lifecycle of stars with masses like our Sun.

- Press the **Information** tab at the top right. Read the overview of the lifecycle stages for a Sun-like star.
- When you are done reading, click the "x" to close the tab.
- Press the start animation button to see a very fast evolution of the Sun. When you are done, you can press the reset button to return the simulation to the start.
- Now select **Normal** speed and run the animation again.
- At the top of the simulation, the name of the current stage will be displayed.

1) At what stage is the star the largest? Smallest?

   largest : Asymptotic Giant Branch Smallest : Oxygen white dwarf

2) Select the temperature control and rerun the animation. At what stage does the temperature increase dramatically? Why is this?

   Asymptotic Giant Branch, the helium core is contracting at a fast rate causing it to heat up

3) At what stage does the temperature decrease dramatically? Why is this?

   White Dwarf stage, all the hydrogen and helium has been cast out by the nebula

4) What happens to the temperature during the red giant phase?

   It increases

5) Select the luminosity control and rerun the animation. At what stage(s) is the luminosity the highest?

   thermal pulsing Asymptotic giant branch

6) At what stage is the luminosity constant?

   <script>

7) At what stage does the luminosity decrease dramatically? Why?

   the white dwarf stage, the star is lossing all its heat and cooling down cause it to not burn as bright like it did before

8) Select the mass control and rerun the animation. How much mass is left after the final stage?

.536

9) Click the Data Table tab at the top. How many stages does a solar-mass star go through? Which two take the least amount of time?

six stages. thermal pulsing and the asymptotic giant branch take the least amount of time

10) Other than the Carbon/Oxygen White Dwarf phase, add up the values in the Duration column of the Data Table. How long did the process take?

11,995 myrs

Low Mass Star

Now, we will look at a low mass star. Reset the simulation and click the **Mass** dropdown menu to select a 0.65 solar mass star.

- Click the **Information** tab at the top and read through the stages of the low mass star. Click the "x" to close the Information tab and return to the simulation.
- Press reset to return the simulation to the start.
- Select **Fast** speed and run the animation.

11) Does it go through the same stages as the Sun?

yes

12) Select the temperature control and rerun the animation. At what stage does the temperature increase dramatically? Is this the same as the Sun?

thermal pulsing asymptotic giant branch, it is the same as the sun

13) Select the luminosity control and rerun the animation. At what stage(s) is the luminosity the highest? How does this compare with the solar mass star?

thermal pulsing asymptotic giant branch, it is the same.

14) Select the mass control and rerun the animation. How much mass is left after the final stage?

.521

15) Click the Data Table tab at the top. How many stages does this mass star go through? Which one takes the least time? How does this compare to the data from the solar mass star?

seven, the solar goes through six

16) Other than the Helium White Dwarf phase, add up the values in the Duration column of the Data Table. How long did the process take? How does this compare to a solar mass star?

38,788 myrs, this one takes much longer

6 Solar Mass Star

Reset the simulation and click the **Mass** dropdown menu to select a 6 solar mass star.

- Click the **Information** tab at the top and read through the stages of the low mass star. Click the "x" to close the Information tab and return to the simulation.
- Press reset to return the simulation to the start.
- Select **Normal** speed and run the animation.

17) How does the lifecycle compare to the solar mass star?

its higher in the main sequence and it moves more than the solar mass star

18) Select the temperature control and rerun the animation. At what stage does the temperature increase dramatically? Is the same as the Sun?

White dwarf stage, it is not the same

19) How is the temperature profile different than that of the solar mass star?

it is very hot while the suns was not

20) Select the mass control and rerun the animation. What is the remaining mass when the star is a white dwarf? How does this mass loss compare with smaller stars like the Sun?

1.151, it is still bigger by double

21) Other than the Carbon/Oxygen White Dwarf phase, ddd up the values in the Duration column of the Data Table. How long did the process take? How does this compare to a solar mass star?

77.75, it is much quicker

20 Solar Mass Star

Reset the simulation and click the **Mass** dropdown menu to select a 20 solar mass star.

- Click the **Information** tab at the top and read through the stages of the low mass star. Click the "x" to close the Information tab and return to the simulation.
- Press reset to return the simulation to the start.
- Select **Normal** speed and run the animation.

22) Are the stages similar to the 6-solar-mass star?

yes

23) At what stage is it the largest?

Core helium burning

24) Select the temperature control and rerun the animation. At what stage is the temperature the highest?

Main Sequence

25) What happens to the temperature as the star ages? Why is this?

It cools, gravity is counteracting it making it cool down

26) What happens to the temperature during the Asymptotic Giant Branch phase?

It stays the same

27) Select the luminosity control and rerun the animation. How does the luminosity change during most of the post-main-sequence stages?

Its stays the same brightness but changes from blue to red

28) Select the mass control and rerun the animation. Does the mass change very much through the star's life?

it changes rapidly once it leaves the main sequence losing most of its mass

29) Click the Data Table tab at the top. How many stages does the star go through? Which stage(s) is the shortest? How does this compare to the solar mass star?

four stages, the asymptotic giant branch is the shortest, the solar mass star it is bigger

30) How does the star ultimately end?

it explodes and becomes a dwarf

31) What is the mass of the final object? How does this mass loss compare to the other stars you observed in the simulation?

2.332, it losses much more mass but still ends up with more mass then smaller stars

32) Other than the Neutron Star phase, how long did the process take? How does this compare to the other stars you observed in this simulation?

the process was much quicker than other stars.